Twitter was 50 percent more volatile than bitcoin in 2016, and bitcoin was nearly three times more volatile than AT&T. The latter is to be expected given bitcoin's network value is less than 5 percent that of AT&T's market cap, and it has been around for less than a decade, while AT&T has been around for more than a century.

In examining FANG stocks, we see an interesting pattern with volatility. Remembering our discussion of modern portfolio theory, historically the most volatile assets have generally been the ones with the greatest returns. This relationship between risk (i.e., volatility) and reward is to be expected: no reward without the accompanying risk. In Figure 7.13 we see that bitcoin's volatility has been the highest, with Netflix coming in second; and these two assets were the best performing. Interestingly, in this period bitcoin's annual returns of 212 percent were threefold that of Netflix's 73 percent, yet bitcoin's volatility was only 35 percent greater than Netflix. Intuitively, it appears bitcoin has had better risk-reward characteristics than Netflix. Similarly, Google, which performed the least well of the FANG stocks with 23 percent returns, also had the lowest volatility at 1.5 percent.

As we learned in the preceding chapter, it's easy enough to directly calculate the risk-reward ratio of different assets. It would appear that in this time period (Facebook's IPO to January 3, 2017) bitcoin has had the best risk-reward ratio of all these assets.

But to make sure, we'll crunch the numbers.